RETAIL SALES &
INVENTORY INTELLIGENCE SYSTEM

1,615	$7.69M	1,445	3	42%
Orders	Revenue	Customers	Stores	YoY Growth

Domain: Retail Sales & Inventory Analytics | Tools: Excel · MySQL · Power BI

Prepared by: Abhijit Sinha

Problem Statement & Business Objectives

PROBLEM STATEMENT

A retail company **operating 3 stores across NY, CA & TX** had no structured analytics system. Management could not identify top-selling products, understand customer behaviour, monitor staff performance, track inventory shortages, or analyse shipment delays. All data sat in raw CSV files with zero visibility.

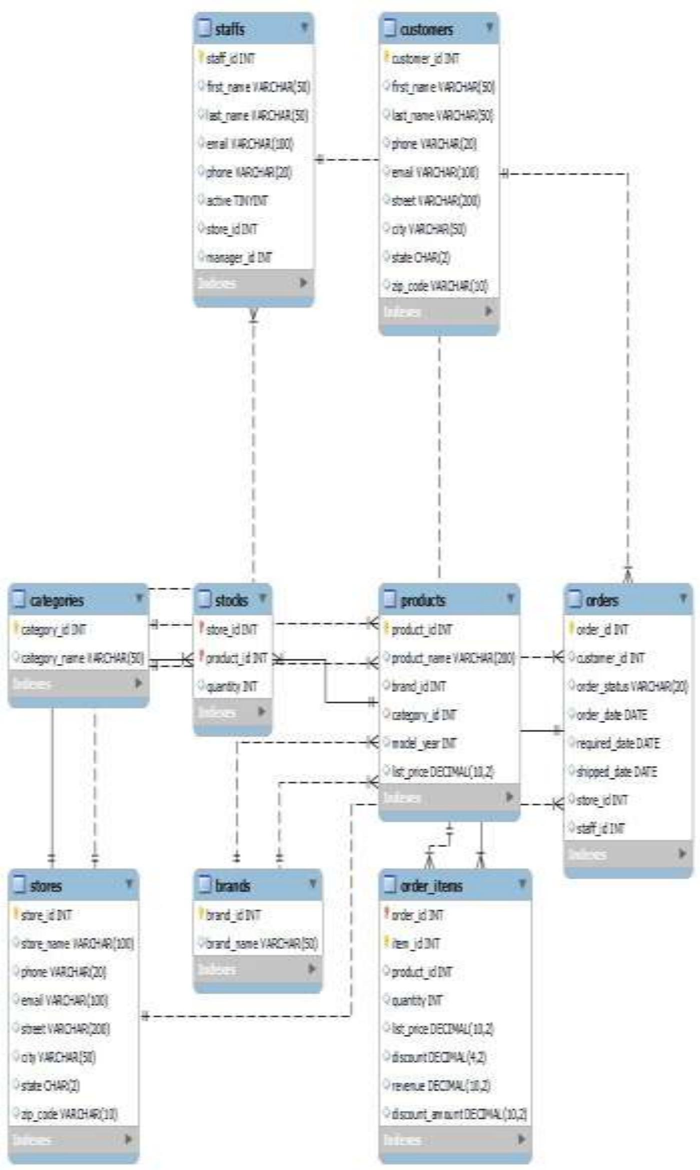
BUSINESS USE CASES

1. Identify top-selling brands by region & store
2. Track customer orders and fulfillment status
3. Evaluate staff performance by total sales handled
4. Identify most profitable product categories
5. Analyse stock levels across stores to optimise inventory
6. Monitor delayed shipments vs required delivery dates
7. Discover customer concentration & buying patterns
8. Understand order trends — daily, weekly, monthly

Dataset Overview — 9 Relational Tables

Table	Rows	Key Information	Domain
orders	1,615	Order transactions, status, dates, store, staff	Sales
order_items	4,722	Products per order, price, discount, revenue	Sales
customers	1,445	Contact details, city, state — NY/CA/TX	Sales
staffs	10	6 active staff across 3 stores	Sales
stores	3	Santa Cruz CA, Baldwin NY, Rowlett TX	Sales
products	321	Bike models, brand, category, price (2016-2018)	Production
brands	9	Trek, Electra, Surly, Sun Bicycles, Haro...	Production
categories	7	Mountain, Road, Cruisers, Electric, Cyclocross...	Production
stocks	939	Inventory quantity per store per product	Production

ER DIAGRAM



Phase 1 — Excel Data Preprocessing

DATA CLEANING STEPS

1

Data Audit

Ran COUNTBLANK() and COUNTIF() on all 9 tables. Found 1,267 null phone values in customers and 170 null shipped_dates in orders stored as text NULL.

2

Null Handling

Phone column: replaced all NULLs with 'Not Provided'. Shipped_date kept blank intentionally — represents orders not yet shipped (status 1,2,3).

3

Duplicate Removal

Identified 30 duplicate product names with different product_ids. Flagged as data quality observation. All 321 rows retained in SQL to preserve FK integrity.

4

Date Standardization

Converted order_date, required_date, shipped_date to YYYY-MM-DD text format using =TEXT(D2,"YYYY-MM-DD") to prevent Excel from reformatting on save.

5

Calculated Columns

Added Revenue = quantity × list_price × (1 − discount) and Discount_Amount in order_items. Added Status_Label and Shipment_Flag (Late/On Time/Not Shipped) in orders.

6

Output

Saved retail_cleaned.xlsx with all 9 cleaned sheets + individual clean_*.csv files for SQL import.

Key Findings: Total Revenue = $7,689,116 | Total Discounts = $889,872 | 458 Late Shipments identified | 170 Unshipped orders flagged

Phase 2 — SQL Database Design & Querying

DATABASE SCHEMA

9 Tables | 8 Foreign Key Relationships | All FK checks passed ✓

brands	brand_id PK, brand_name
categories	category_id PK, category_name
stores	store_id PK, store_name, city, state
customers	customer_id PK, name, email, state
staffs	staff_id PK, name, store_id FK, manager_id
products	product_id PK, name, brand_id FK, category_id FK
orders	order_id PK, customer_id FK, store_id FK, staff_id FK
order_items	order_id FK + item_id PK, product_id FK, qty, price
stocks	store_id FK + product_id FK (composite PK), qty

SQL QUERIES WRITTEN

Total Sales Summary
Revenue, orders, customers, discounts in one query

Revenue by Store & State
Baldwin NY $5.22M (67.8%) · Santa Cruz CA · Rowlett TX

Revenue by Brand
Trek $4.60M (59.9%) · Electra $1.21M · Surly $0.95M

Revenue by Category
Mountain Bikes 35.3% · Road Bikes 21.7%

Staff Performance
Boyer $2.62M · Daniel $2.59M — top 2 handle 68%

Late Shipment Analysis
31.7% shipped late — Santa Cruz worst at 36.6%

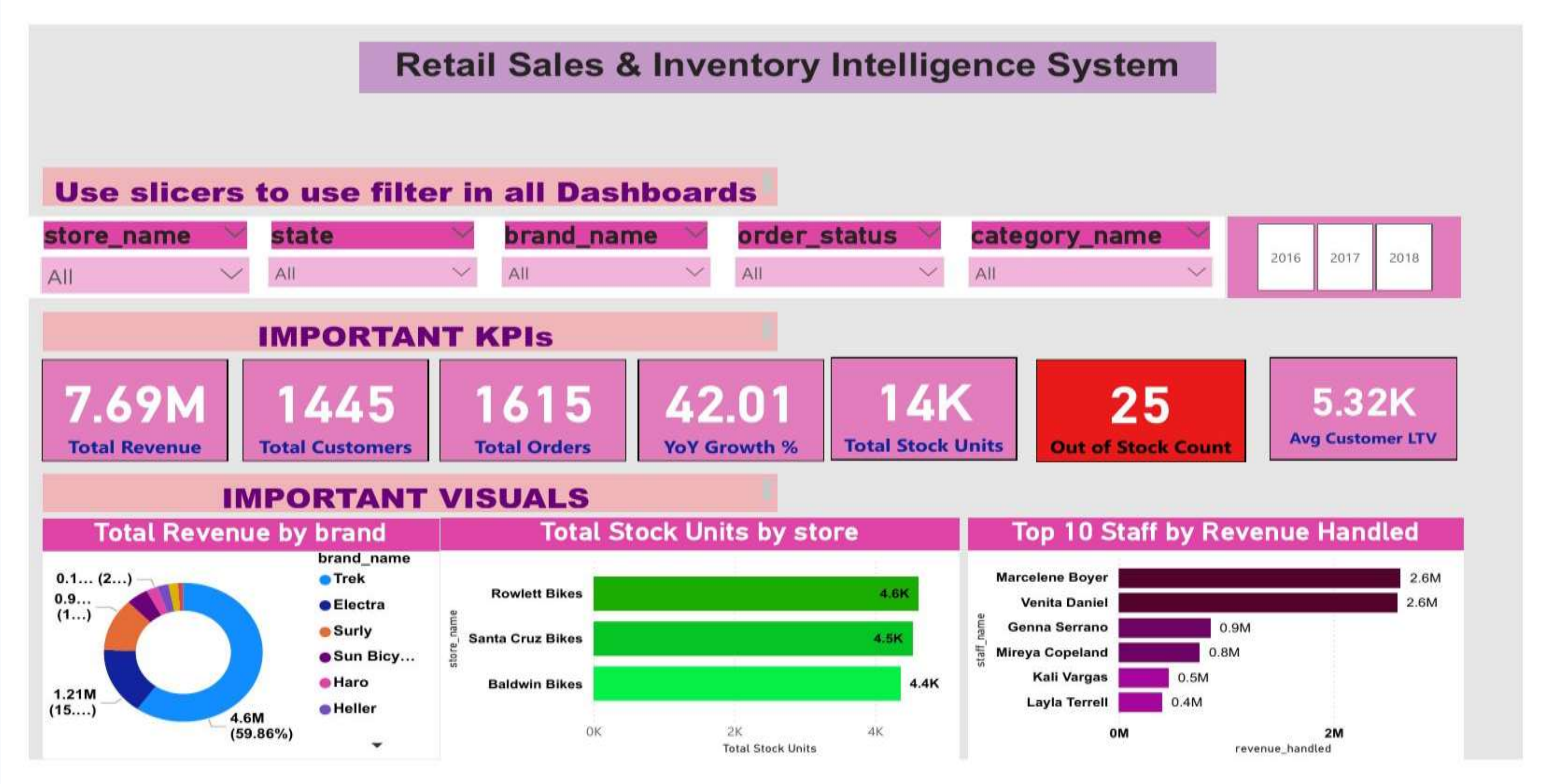
Customer Loyalty
91% one-time buyers — only 9% repeat customers

Year-over-Year Trend
2016 $2.43M → 2017 $3.45M (+42%) → 2018 partial

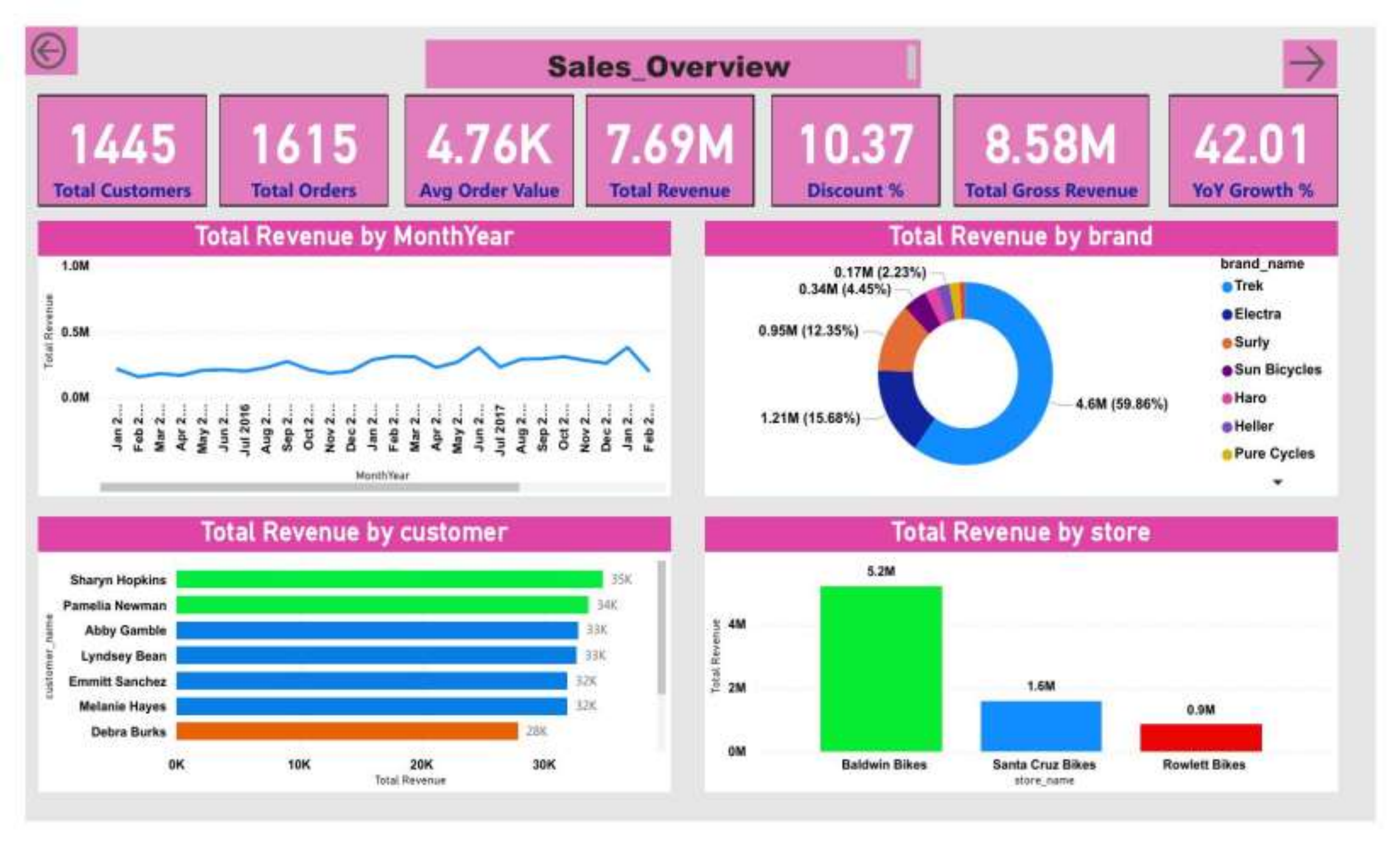
Inventory Alerts
25 out of stock · 141 low stock items flagged

4 SQL Views created for Power BI: vw_sales_master · vw_stock_summary · vw_staff_performance · vw_customer_ltv

Phase 3 — Power BI Dashboard: Executive Summary



Phase 3 — Power BI Dashboard: Sales Overview



$7.69M
Total Revenue

1,615
Orders

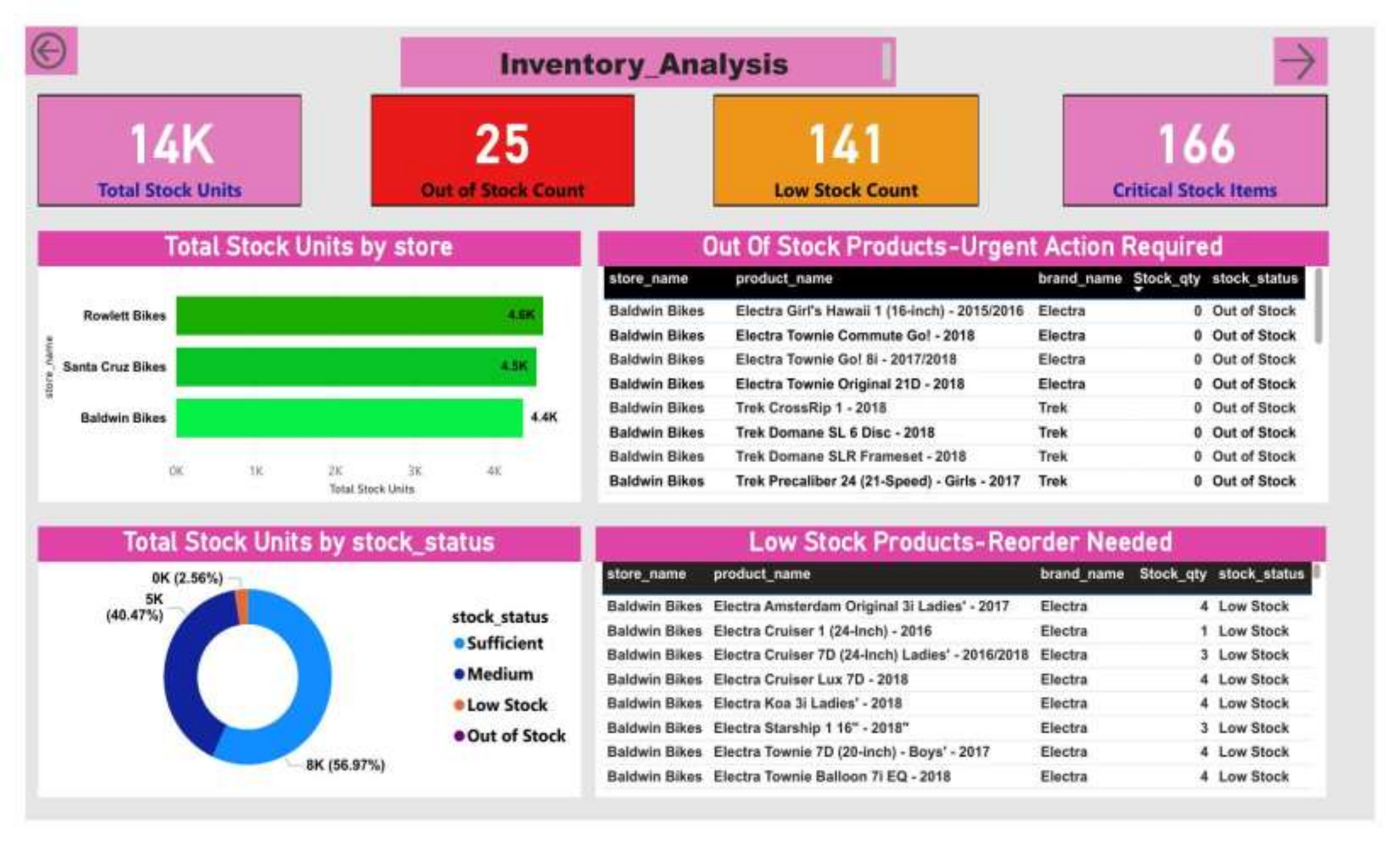
42.01%
YoY Growth

10.37%
Discount Rate

Trek
Top Brand (59.9%)

Baldwin
Top Store (67.8%)

Phase 3 — Power BI Dashboard: Inventory Analysis



14K
Total Stock Units

25
Out of Stock

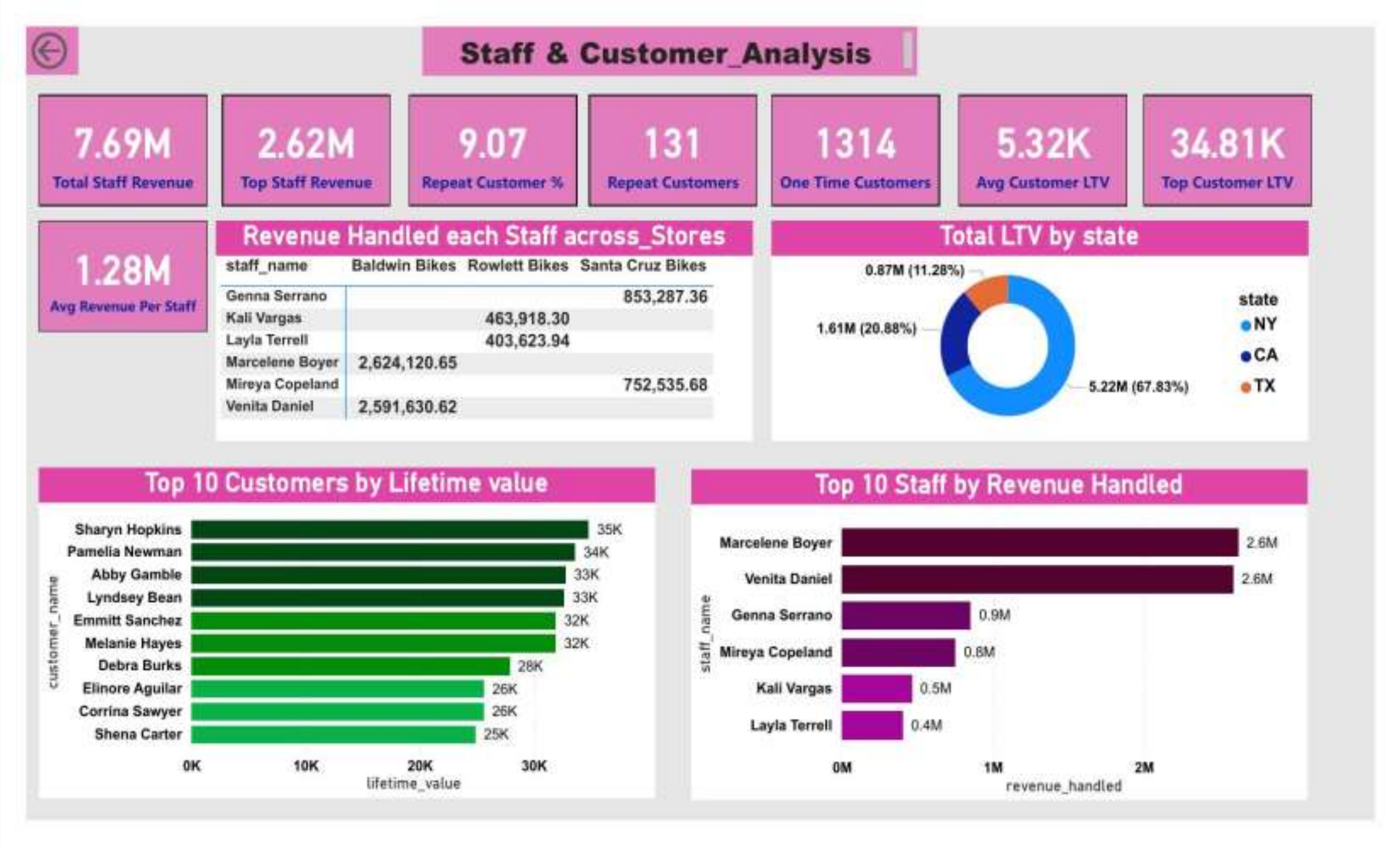
141
Low Stock Items

166
Critical Items

855
Rows Imported

3
Stores Tracked

Phase 3 — Power BI Dashboard: Staff & Customer Analysis



$2.62M
Top Staff Revenue

9.07%
Repeat Buyers

$34.8K
Top Customer LTV

$5.32K
Avg Customer LTV

1,314
One-Time Customers

NY
70% of Customers

Key Business Insights

Findings from your actual retail data — 2016 to 2018

1 Store Concentration Risk

Baldwin Bikes (NY) generates 67.8% of all revenue ($5.22M of $7.69M). Extreme single-store dependency — any disruption eliminates two-thirds of income.

2 Trek Brand Dominance

Trek alone accounts for 59.9% of revenue ($4.60M). Top product: Trek Slash 8 27.5 at $555,559. Heavy single-brand reliance creates supply chain risk.

3 Mountain Bikes Lead Revenue

Mountain Bikes: $2.72M (35.3%) + Road Bikes: $1.67M (21.7%) = 57% of all revenue. These two categories should be prioritised for stocking and promotions.

4 Critical Late Shipment Problem

31.7% of shipped orders (458 of 1,445) arrived after the required date. Santa Cruz Bikes worst at 36.6%. All 3 stores exceed acceptable thresholds.

5 Customer Retention Crisis

91% of 1,445 customers placed only 1 order. Only 131 customers (9%) are repeat buyers. Converting 5% more to repeat buyers would add ~$430K in annual revenue.

6 Strong 2017 Growth (+42%)

Revenue grew from $2.43M (2016) to $3.45M (2017) — 42% growth. 2018 data ends in April showing $1.81M. 2018 is a partial-year dataset.

Business Recommendations

5 actionable recommendations based on data analysis findings

R1

Fix Late Shipments Urgently

Current: 31.7% late rate across all stores

Action:
Audit fulfilment process at all 3 stores. Set KPI: below 10% late within 6 months. Investigate Santa Cruz Bikes first (36.6% late). Implement daily shipment tracking dashboard.

R2

Launch Customer Retention Programme

Current: 91% customers never return

Action:
Introduce post-purchase email follow-ups, loyalty scheme, and targeted promotions for 1,314 one-time buyers. Even 5% conversion adds ~$430K annual revenue.

R3

Reduce Store & Brand Dependency

Current: Baldwin 67.8%, Trek 59.9%

Action:
Invest in marketing and staffing for Santa Cruz CA and Rowlett TX. Negotiate with 2-3 additional brands to reduce Trek reliance below 50%.

R4

Automate Inventory Reorder Alerts

Current: 25 out-of-stock, 141 low-stock products

Action:
Implement auto reorder triggers at 10-unit threshold. Prioritise Trek Mountain Bikes and Road Bikes. Assign store manager accountability for stock levels.

R5

Review & Optimise Discount Strategy

Current: $889,872 discounts given (10.5% avg)

Action:
Audit the 20% discount tier applied to 1,203 orders. Test reducing to 15% and measure conversion impact. Target discounts at slow-moving inventory only.

Future Scope

Building on this foundation to deliver advanced analytics capabilities

01

ML Demand Forecasting

Use Python Scikit-learn to predict high-demand products by month. Train on 2016-2017 data, validate in 2018. Enable proactive inventory management before stock runs out.

02

Customer RFM Segmentation

Cluster 1,445 customers into Champion, Loyal, At-Risk, and Lost segments using KMeans. Enable targeted marketing with personalised campaigns for each segment.

03

Real-time Data Pipeline

Replace manual CSV import with an automated ETL pipeline (Apache Airflow or Azure Data Factory) that refreshes the Power BI dashboard daily from live POS transaction data.

04

Churn Prediction Model

Build a binary classification model (Logistic Regression / Random Forest) to identify customers likely to not return. Allow the business to intervene with targeted offers proactively.

05

Supplier & Shipment Intelligence

Extend analysis to track supplier lead times, carrier performance, and delivery reliability. Directly address the 31.7% late shipment problem at its root cause.

06

Mobile-First Dashboard

Publish Power BI reports to Power BI Service and optimise mobile layouts. Allow store managers to monitor KPIs on smartphones in real time from the shop floor.

Thank You

Retail Sales & Inventory Intelligence System — Project Summary

Excel	SQL	Power BI	Insights
9 Sheets Cleaned 30 Duplicates Flagged 5 Columns Added	9 Tables Created 9 Queries Written 4 Views Built	3 Dashboards 1 Executive Page 32 DAX Measures 6 Synced Slicers	6 Key Findings 5 Recommendations 6 Future Scope Items

Project Deliverables Submitted:

- Retail_Cleaned.xlsx
- Retail_Analysis.sql
- Retail_Analysis_Dashboard.pbix
- ER_Diagram.png
- Project_Report.pptx